

Assignment 5

April 23, 2026

Question 1. (50 points) *In class, we covered the Markowitz Mean–Variance Portfolio Optimization problem using historical data from Yahoo Finance. In the in-class example, we formulated the model as a risk-minimization problem subject to achieving a target return.*

*In this question, your task is to formulate and apply the **reverse** version of this model: one that maximizes expected return subject to staying below a specified risk threshold.*

1. (10 points) *Write the complete algebraic formulation of this model. Make sure to clearly define all variables, parameters, objective, and constraints.*
2. (15 points) *Using historical daily price data for all S&P 500 stocks over the three-year period prior to the start of this assignment, construct a portfolio that satisfies your formulation. Present a visualization of your portfolio that clearly shows the percentage allocation to each asset (e.g., bar chart or pie chart), and provide a brief explanation of your results.*
3. (15 points) *Perform a sensitivity analysis of the return–risk trade-off using the efficient frontier. Provide a short explanation (2–3 sentences) summarizing your insights.*
4. (10 points) *In practice, the classical Markowitz model often produces portfolios with many small, non-zero allocations, which may be difficult to interpret or implement.*

Modify your model by introducing simple additional constraints or penalty that improves the practicality of the portfolio. You may choose one of the following (or propose your own with justification):

- *Limit the maximum allocation to any single asset (e.g., $w_i \leq 0.1$)*
- *Enforce a minimum allocation threshold for selected assets (e.g., $w_i \geq 0.01$)*
- *Add a sparsity-inducing constraint or penalty to reduce the number of assets held*

Solve the modified model and compare your results with the original portfolio. Briefly discuss (2–4 sentences):

- *How did your portfolio change?*
- *What trade-offs did you observe?*

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Question 2. (25 points) Use the KKT conditions to find the optimal solution to the following convex optimization problem:

$$\min_{x_1, x_2} (x_1 - 2)^2 + (x_2 - 4)^2 \quad (1)$$

$$s.t. \quad -x_1 + x_2 = 1 \quad (2)$$

$$x_1 + x_2 \leq 4 \quad (3)$$

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Question 3. (25 points) In a transportation network, sending flow along a road causes congestion. The congestion cost on edge (i,j) is given by

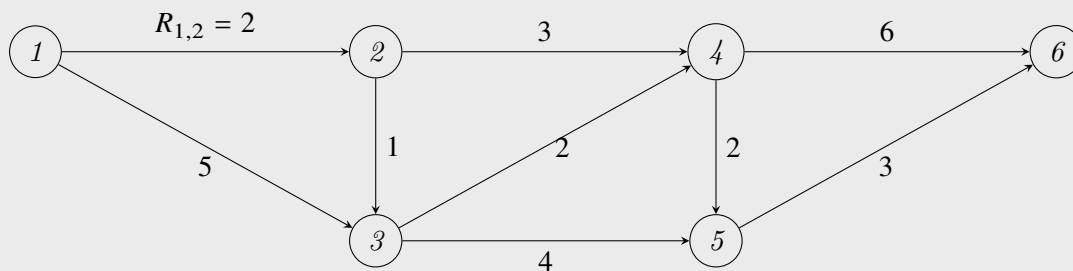
$$R_{ij}x_{ij}^2,$$

where x_{ij} is the number of vehicles traveling on road (i,j) and R_{ij} is a road-specific congestion parameter.

Consider the network below. A total of **150 vehicles** must be routed from node 1 (origin) to node 6 (destination). The congestion parameter R_{ij} for each road is shown on the network.

Flow must satisfy **conservation of flow**:

- Node 1 supplies 150 vehicles
- Node 6 demands 150 vehicles
- All intermediate nodes have net flow equal to 0



1. (15 points) Formulate a **quadratic program (QP)** that determines the flow x_{ij} on each road that minimizes the total congestion cost in the network.

Clearly define:

- Decision variables
- Objective function
- Constraints

2. (10 points) Write out a **.lp** file and use Gurobi to solve your formulation.

- Report the optimal flow on each edge
- Report the total minimized congestion cost